Basic Image Operations

Images as functions

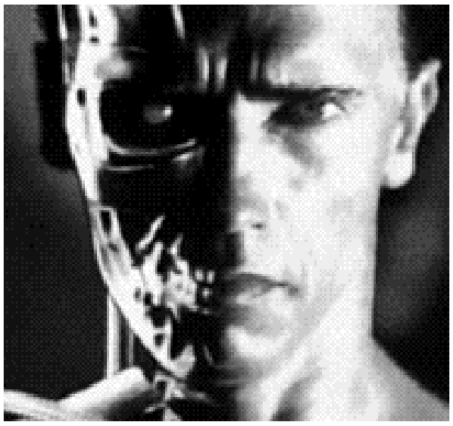
We can think of an **image** as a function, *f*, from $\mathbb{R}^2$ to $\mathbb{R}$:

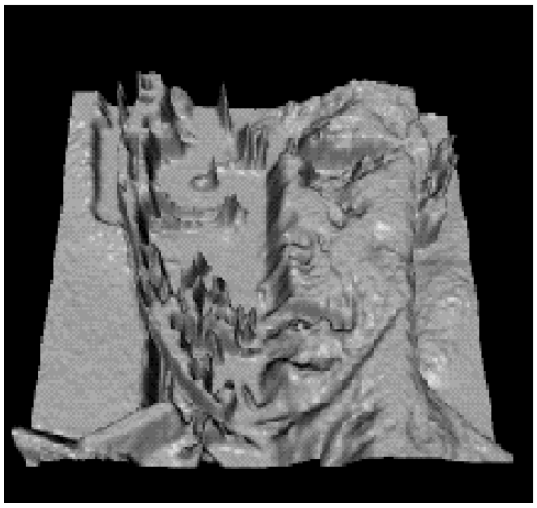
- $f(x, y)$ gives the **intensity** at position (x, y)
- Realistically, we expect the image only to be defined over a rectangle, with a finite range:
  - $f: [a,b] \times [c,d] \rightarrow [0,1]$

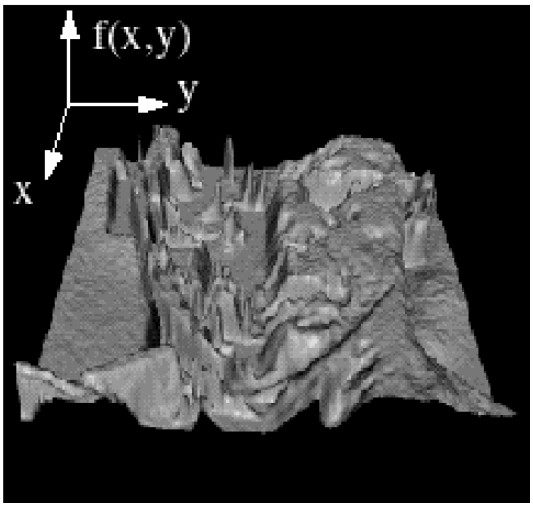
A color image is just three functions pasted together. We can write this as a "vector-valued" function:

$$f(x, y) = \begin{bmatrix} r(x, y) \\ g(x, y) \\ b(x, y) \end{bmatrix}$$

Images as functions







What is a digital image?

We usually work with **digital** (**discrete**) images:

- **Sample** the 2D space on a regular grid
- **Quantize** each sample (round to nearest integer)

If our samples are Δ apart, we can write this as:

$$f[i, j] = \text{Quantize}\{ f(i\,\Delta, j\,\Delta) \}$$

The image can now be represented as a matrix of integer values

$j \rightarrow$

$i \downarrow$

62	79	23	119	120	105	4	0
10	10	9	62	12	78	34	0
10	58	197	46	46	0	0	48
176	135	5	188	191	68	0	49
2	1	1	29	26	37	0	77
0	89	144	147	187	102	62	208
255	252	0	166	123	62	0	31
166	63	127	17	1	0	99	30

Noise Removal - Filtering

Let's write this down as an equation. Assume the averaging window is (2k+1)x(2k+1):

$$G[i,j] = \frac{1}{(2k+1)^2} \sum_{u=-k}^{k} \sum_{v=-k}^{k} F[i+u, j+v]$$

We can generalize this idea by allowing different weights for different neighboring pixels:

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v] F[i+u, j+v]$$

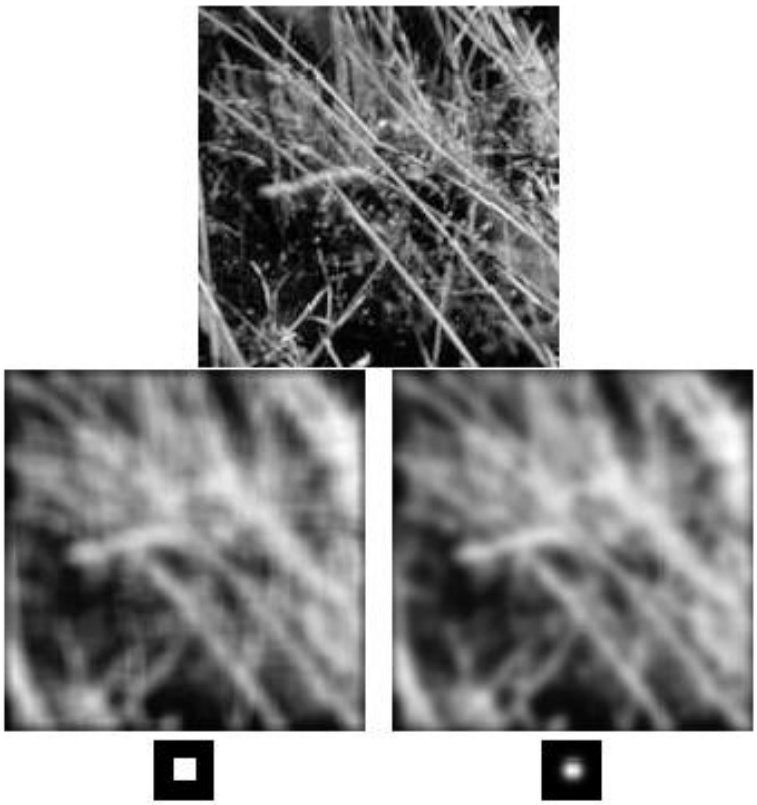
This is called a **cross-correlation** operation and written:

$$G = H \otimes F$$

H is called the "filter," "kernel," or "mask."

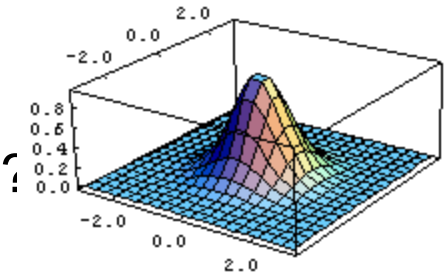
The above allows negative filter indices. When you implement need to use: H[u+k,v+k] instead of H[u,v]

Mean vs Gaussian filtering



This kernel is an approximation of a Gaussian function:

$$h(u,v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{\sigma^2}}$$



What happens if you increase σ ?

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Convolution

A **convolution** operation is a cross-correlation where the filter is flipped both horizontally and vertically before being applied to the image:

$$G[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} H[u,v]F[i-u,j-v]$$

It is written: $$G = H \star F$$

How does convolution differ from cross-correlation?

Suppose F is an impulse function. What will G look like?

Image gradient

How can we differentiate a *digital* image F[x,y]?

- Option 1: reconstruct a continuous image, *f,* then take gradient
- Option 2: take discrete derivative (finite difference)

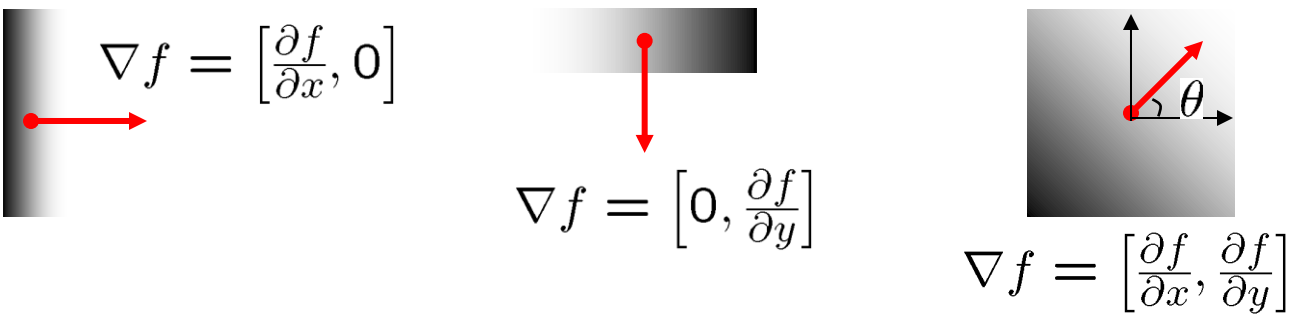
$$\frac{\partial f}{\partial x}[x, y] \approx F[x+1, y] - F[x, y]$$

How would you implement this as a cross-correlation?

Image gradient

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

It points in the direction of most rapid change in intensity



The gradient direction is given by:

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

- how does this relate to the direction of the edge?

The *edge strength* is given by the gradient magnitude

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

original

smoothed (5x5 Gaussian)

smoothed – original

(scaled by 4, offset +128)

Why does this work?

Continuous Filters

We can also apply filters to *continuous* images.

In the case of cross correlation: $g = h \otimes f$

$$g(x,y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u,v) f(x+u, y+v) du dv$$

In the case of convolution: $g = h \star f$

$$g(x,y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u,v) f(x-u, y-v) du dv$$